

04SLN3

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Problem

Find all functions $f : \mathbb{N} \rightarrow \mathbb{N}$ satisfying

$$f(m)^2 + f(n) \mid (m^2 + n)^2$$

for any two positive integers m and n .

Solution. Let $P(m, n)$ be the assertion.

$P(1, 1) \implies f(1)^2 + f(1) \mid 4$. So There are 3 cases,

(i) $f(1)^2 + f(1) = 1$

(ii) $f(1)^2 + f(1) = 2$

(iii) $f(1)^2 + f(1) = 4$.

Of course (i) and (iii) does not give any solutions in integers and $f(1)^2 + f(1) = 2$ is the only equation what works which gives $f(1) = 1$.

$$P(1, p-1) \implies 1 + f(p-1) \mid p^2 \implies f(p-1) \in \{p-1, p^2-1\}.$$

$P(p-1, 1) \implies p^4 - 2p^2 + 2 \mid p^4 - 4p^3 + 8p^2 - 8p + 4$, for $p \neq 2$, $\implies p^4 - 2p^2 + 2 \mid -2p^3 + 5p^2 - 4p + 1$ which can be easily contradicted due to size reasons for large enough p .

As a result for large enough p , $f(p-1) = p-1$.

$(n + (p-1)^2)^2 \equiv (f(n) - n)^2 \equiv 0 \pmod{f(n) + (p-1)^2}$. Of course for large enough p , $f(n) + (p-1)^2 > |f(n) - n|$ making $f(n) - n = 0$ for all $n \in \mathbb{N}$. So $f(n) = n$ for all $n \in \mathbb{N}$.